A Gibbs sampler for Optimal Multiple Co-Inertia Analysis

1 Probabilistic model

We consider K column-centered blocks $\mathbf{X}_1, \dots, \mathbf{X}_K$, with $\mathbf{X}_k \in \mathcal{M}_{N,p_k}(\mathbb{R})$, and a latent space of dimension $H \leq \min_k p_k$. OMCOA minimizes

$$\sum_{k=1}^{K} \left\| \mathbf{X}_k - \mathbf{U}\Sigma_k \mathbf{V}_k^\intercal \right\|^2 \quad \text{subject to} \quad \mathbf{U}^\intercal\mathbf{U} = \mathbf{I}_H, \quad \mathbf{V}_k^\intercal\mathbf{V}_k = \mathbf{I}_H, \quad \Sigma_k = \operatorname{diag}\big(\sigma_k^{(1)}, \dots, \sigma_k^{(H)}\big). \tag{1}$$

A probabilistic counterpart of (1) is obtained by assuming a Gaussian noise:

$$\mathbf{X}_k = \mathbf{U}\Sigma_k \mathbf{V}_k^\intercal + \mathbf{E}_k, \qquad \mathbf{E}_k \text{ with i.i.d. entries } \mathcal{N}\big(0, \tau_k^{-1}\big), \qquad k = 1, \dots, K, \tag{2}$$

where τ_k is the noise precision of block k. Let $\mathcal{V}_{m,H} = \{\mathbf{W} \in \mathcal{M}_{m,H}(\mathbb{R}) : \mathbf{W}^\intercal\mathbf{W} = \mathbf{I}_H\}$ denote the Stiefel manifold. Since the blocks are centered, the latent components are constrained to be centered as well: $\mathbf{U} \in \mathcal{V}_{N,H}$ with $\mathbf{1}_N^\intercal \mathbf{U} = 0$.

Priors. We use the following independent priors:

- $\mathbf{U}$ and $\mathbf{V}_1, \dots, \mathbf{V}_K$ are uniformly distributed on their Stiefel manifolds (non-informative);
- $\sigma_k^{(h)} \sim \mathcal{N}^+\big(0, \lambda^{-1}\big)$, a half-normal distribution with a small precision λ (e.g. $\lambda = 10^{-2}$); the positivity constraint removes the sign indeterminacy between $\sigma_k^{(h)}$ and $\mathbf{v}_k^{(h)}$;
- $\tau_k \sim \text{Gamma}(a_0, b_0)$ with small hyperparameters (e.g. $a_0 = b_0 = 10^{-3}$).

2 Full conditional distributions

Likelihood. Because $\mathbf{U}$ and $\mathbf{V}_k$ have orthonormal columns and Σ_k is diagonal,

$$\left\| \mathbf{X}_k - \mathbf{U}\Sigma_k \mathbf{V}_k^\intercal \right\|^2 = \|\mathbf{X}_k\|^2 - 2\,\operatorname{trace}\big(\Sigma_k \mathbf{V}_k^\intercal \mathbf{X}_k^\intercal \mathbf{U}\big) + \sum_{h=1}^{H} \big(\sigma_k^{(h)}\big)^2. \tag{3}$$

Since the blocks are centered, each $\mathbf{X}_k$ carries $(N-1)p_k$ degrees of freedom, and the log-likelihood reads

$$\log p(\mathbf{X}_1, \dots, \mathbf{X}_K \mid \cdot) = \sum_{k=1}^{K} \Big[\tfrac{(N-1)p_k}{2} \log \tau_k - \tfrac{\tau_k}{2} \left\| \mathbf{X}_k - \mathbf{U}\Sigma_k \mathbf{V}_k^\intercal \right\|^2 \Big] + \text{const.}$$

By (3), $\mathbf{U}$ and $\mathbf{V}_k$ only appear in the linear term $\operatorname{trace}(\Sigma_k \mathbf{V}_k^\intercal \mathbf{X}_k^\intercal \mathbf{U})$. This is the key property: all the full conditionals belong to standard families.

Matrix von Mises–Fisher distribution. A random matrix $\mathbf{W} \in \mathcal{V}_{m,H}$ follows the matrix von Mises–Fisher distribution vMF($\mathbf{F}$), with parameter $\mathbf{F} \in \mathcal{M}_{m,H}(\mathbb{R})$, if its density with respect to the uniform measure on $\mathcal{V}_{m,H}$ is proportional to $\exp\{\operatorname{trace}(\mathbf{F}^\intercal \mathbf{W})\}$ [3]. If $\mathbf{F} = \mathbf{A}\mathbf{D}\mathbf{B}^\intercal$ is the (thin) SVD of $\mathbf{F}$, the mode of vMF($\mathbf{F}$) is the polar factor $\mathbf{A}\mathbf{B}^\intercal$, and the distribution concentrates around it as the singular values of $\mathbf{F}$ increase.

Proposition 1 (Full conditionals)**.** *Under model* (2) *and the above priors:*

1. ***Common space.*** *Let $\mathbf{Q} \in \mathcal{M}_{N,N-1}(\mathbb{R})$ be an orthonormal basis of $\mathbf{1}_N^\perp$ and write $\mathbf{U} = \mathbf{Q}\tilde{\mathbf{U}}$ with $\tilde{\mathbf{U}} \in \mathcal{V}_{N-1,H}$. Then*

$$\tilde{\mathbf{U}} \mid \cdot \;\sim\; \mathrm{vMF}\big(\mathbf{Q}^\intercal \mathbf{F}_{\mathbf{U}}\big), \qquad \mathbf{F}_{\mathbf{U}} = \sum_{k=1}^{K} \tau_k \, \mathbf{X}_k \mathbf{V}_k \Sigma_k.$$

2. ***Block loadings.*** *For $k = 1, \dots, K$,*

$$\mathbf{V}_k \mid \cdot \;\sim\; \mathrm{vMF}\big(\mathbf{F}_{\mathbf{V}_k}\big), \qquad \mathbf{F}_{\mathbf{V}_k} = \tau_k \, \mathbf{X}_k^\intercal \mathbf{U} \Sigma_k.$$

3. ***Block weights.*** *For $k = 1, \dots, K$ and $h = 1, \dots, H$, let $d_k^{(h)} = \mathbf{u}^{(h)\intercal} \mathbf{X}_k \mathbf{v}_k^{(h)}$. Then*

$$\sigma_k^{(h)} \mid \cdot \;\sim\; \mathcal{N}^{+}\Big(\frac{\tau_k \, d_k^{(h)}}{\tau_k + \lambda}, \; \frac{1}{\tau_k + \lambda}\Big),$$

   a normal distribution truncated to $(0, +\infty)$.

4. ***Noise precisions.*** *For $k = 1, \dots, K$,*

$$\tau_k \mid \cdot \;\sim\; \mathrm{Gamma}\Big(a_0 + \frac{(N-1)p_k}{2}, \; b_0 + \frac{1}{2}\big\|\mathbf{X}_k - \mathbf{U}\Sigma_k \mathbf{V}_k^\intercal\big\|^2\Big).$$

Sketch of proof. (1)–(2): by (3), the conditional density of $\mathbf{U}$ is proportional to $\exp\{\sum_k \tau_k \operatorname{trace}(\Sigma_k \mathbf{V}_k^\intercal \mathbf{X}_k^\intercal \mathbf{U})\} = \exp\{\operatorname{trace}(\mathbf{F}_{\mathbf{U}}^\intercal \mathbf{U})\}$, and $\operatorname{trace}(\mathbf{F}_{\mathbf{U}}^\intercal \mathbf{Q}\tilde{\mathbf{U}}) = \operatorname{trace}\big((\mathbf{Q}^\intercal \mathbf{F}_{\mathbf{U}})^\intercal \tilde{\mathbf{U}}\big)$; the same argument applies to $\mathbf{V}_k$ with $\operatorname{trace}(\Sigma_k \mathbf{V}_k^\intercal \mathbf{X}_k^\intercal \mathbf{U}) = \operatorname{trace}(\mathbf{F}_{\mathbf{V}_k}^\intercal \mathbf{V}_k)/\tau_k$. (3): the terms of (3) involving $\sigma_k^{(h)}$ are $-\frac{\tau_k}{2}\big[(\sigma_k^{(h)})^2 - 2\sigma_k^{(h)} d_k^{(h)}\big]$; adding the log-prior $-\frac{\lambda}{2}(\sigma_k^{(h)})^2$ and completing the square gives a Gaussian kernel, restricted to $\sigma_k^{(h)} > 0$. (4): the Gamma prior is conjugate for the precision of a Gaussian likelihood. □

3 Sampling from the matrix von Mises–Fisher distribution

Direct sampling from vMF($\mathbf{F}$) is not straightforward, but its column-wise conditionals are vector von Mises–Fisher distributions [3]. Let $\mathbf{W}_{-j}$ be $\mathbf{W}$ without its j-th column and $\mathbf{N}_j \in \mathcal{M}_{m,m-H+1}(\mathbb{R})$ an orthonormal basis of $\operatorname{span}(\mathbf{W}_{-j})^\perp$. Then

$$\mathbf{w}_j = \mathbf{N}_j \mathbf{z}, \qquad \mathbf{z} \sim \mathrm{vMF}\big(\mathbf{N}_j^\intercal \mathbf{f}_j\big) \text{ on the unit sphere } \mathbb{S}^{m-H},$$

where $\mathbf{f}_j$ is the j-th column of $\mathbf{F}$. One sweep over the H columns (in random order) is a valid Gibbs update of $\mathbf{W}$, which is enough inside the Gibbs sampler. The vector distribution vMF($\mathbf{f}$), with density proportional to $\exp(\mathbf{f}^\intercal \mathbf{z})$ on $\mathbb{S}^{m-1}$, is sampled with the rejection algorithm of Wood [4] (Algorithm 1).

The expression of b used in Algorithm 1 is algebraically equivalent to Wood's original one, $b = \big(-2\kappa + \sqrt{4\kappa^2 + (m-1)^2}\big)/(m-1)$, but avoids a numerical cancellation when κ is large, which is the typical situation here since κ scales with the noise precisions τ_k.

Algorithm 1 Sampling $\mathbf{z} \sim \text{vMF}(\mathbf{f})$ on $\mathbb{S}^{m-1}$ (Wood, 1994)

$\kappa \leftarrow \|\mathbf{f}\|, \quad \boldsymbol{\mu} \leftarrow \mathbf{f}/\kappa$
$b \leftarrow (m-1)\big/\big(2\kappa + \sqrt{4\kappa^2 + (m-1)^2}\big), \quad x_0 \leftarrow (1-b)/(1+b), \quad c \leftarrow \kappa x_0 + (m-1)\log(1-x_0^2)$
repeat
  draw $\zeta \sim \text{Beta}\big(\frac{m-1}{2}, \frac{m-1}{2}\big)$ and $\upsilon \sim \mathcal{U}(0,1)$
  $w \leftarrow \big(1-(1+b)\zeta\big)\big/\big(1-(1-b)\zeta\big)$
until $\kappa w + (m-1)\log(1-x_0 w) - c \geq \log \upsilon$
draw $\boldsymbol{\xi} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_m)$, set $\boldsymbol{\xi} \leftarrow \boldsymbol{\xi} - (\boldsymbol{\mu}^\intercal \boldsymbol{\xi})\boldsymbol{\mu}$ and $\boldsymbol{\xi} \leftarrow \boldsymbol{\xi}/\|\boldsymbol{\xi}\|$
return $\mathbf{z} = w\,\boldsymbol{\mu} + \sqrt{1-w^2}\,\boldsymbol{\xi}$

4 The Gibbs sampler

Algorithm 2 Gibbs sampler for OMCOA

Require: centered blocks $\mathbf{X}_1, \ldots, \mathbf{X}_K$; dimension H; number of iterations T; burn-in T_0; hyperparameters a_0, b_0, λ
Initialization: $\mathbf{U} = \mathbf{Q}\tilde{\mathbf{U}}$, $\mathbf{V}_k$, Σ_k from OMCOA (or MCOA, or at random); $\tau_k \leftarrow (N-1)p_k\big/\|\mathbf{X}_k - \mathbf{U}\Sigma_k\mathbf{V}_k^\intercal\|^2$
for $t = 1, \ldots, T$ **do**
  $\mathbf{F}_\mathbf{U} \leftarrow \sum_k \tau_k\,\mathbf{X}_k\mathbf{V}_k\Sigma_k$; update $\tilde{\mathbf{U}}$ by one column-wise sweep of $\text{vMF}(\mathbf{Q}^\intercal\mathbf{F}_\mathbf{U})$; $\mathbf{U} \leftarrow \mathbf{Q}\tilde{\mathbf{U}}$
  for $k = 1, \ldots, K$ **do**
    $\mathbf{F}_{\mathbf{V}_k} \leftarrow \tau_k\,\mathbf{X}_k^\intercal\mathbf{U}\Sigma_k$; update $\mathbf{V}_k$ by one column-wise sweep of $\text{vMF}(\mathbf{F}_{\mathbf{V}_k})$
    for $h = 1, \ldots, H$ **do**
      $d_k^{(h)} \leftarrow \mathbf{u}^{(h)\intercal}\mathbf{X}_k\mathbf{v}_k^{(h)}$; draw $\sigma_k^{(h)} \sim \mathcal{N}^+\big(\tau_k d_k^{(h)}/(\tau_k+\lambda),\, 1/(\tau_k+\lambda)\big)$
    end for
    draw $\tau_k \sim \text{Gamma}\big(a_0 + (N-1)p_k/2,\; b_0 + \|\mathbf{X}_k - \mathbf{U}\Sigma_k\mathbf{V}_k^\intercal\|^2/2\big)$
  end for
  if $t > T_0$ **then** store $\big(\mathbf{U}, \mathbf{V}_1, \ldots, \mathbf{V}_K, \Sigma_1, \ldots, \Sigma_K, \tau_1, \ldots, \tau_K\big)$
  end if
end for
Post-processing: align each stored draw on a reference solution (Section 5)
return the aligned draws

5 Post-processing: label switching and signs

The posterior distribution is invariant under any permutation π of the H dimensions applied jointly to $\mathbf{U}$, $\mathbf{V}_1, \ldots, \mathbf{V}_K$ and $\Sigma_1, \ldots, \Sigma_K$, and under a joint change of sign of $\mathbf{u}^{(h)}$ and $\mathbf{v}_1^{(h)}, \ldots, \mathbf{v}_K^{(h)}$. Posterior summaries are therefore computed after aligning each draw on a reference $\mathbf{U}_{\text{ref}}$ (in practice the OMCOA solution). For each draw, with $\mathbf{C} = \mathbf{U}_{\text{ref}}^\intercal\mathbf{U}$,

$$\hat{\pi} = \operatorname*{arg\,max}_{\pi \in \mathfrak{S}_H} \sum_{h=1}^{H} \big|C_{h,\pi(h)}\big|, \qquad s_h = \text{sign}\big(C_{h,\hat{\pi}(h)}\big),$$

and the columns of $\mathbf{U}$ and $\mathbf{V}_k$ are reordered by $\hat{\pi}$ and multiplied by s_h, while the diagonal of Σ_k is reordered by $\hat{\pi}$ (it stays positive). Posterior means and credible intervals of $\sigma_k^{(h)}$ are then obtained from the empirical distribution of the aligned draws.

6 Link with OMCOA

Remark 1 (OMCOA as iterated conditional modes)**.** *The modes of the full conditionals of Proposition 1 are*

$$\mathbf{U} = \mathbf{A}\mathbf{B}^\intercal \text{ with } \mathbf{F}_\mathbf{U} = \mathbf{A}\mathbf{D}\mathbf{B}^\intercal, \qquad \mathbf{V}_k = \mathbf{A}_k\mathbf{B}_k^\intercal \text{ with } \mathbf{F}_{\mathbf{V}_k} = \mathbf{A}_k\mathbf{D}_k\mathbf{B}_k^\intercal, \qquad \sigma_k^{(h)} = \frac{\tau_k}{\tau_k + \lambda}\, d_k^{(h)}.$$

When $\lambda \to 0$ and the precisions are equal ($\tau_1 = \cdots = \tau_K$), these are exactly the OMCOA updates: Procrustes rotation of $\mathbf{X}\mathbf{H}^\intercal$ for $\mathbf{U}$ (with $\mathbf{H} = [\Sigma_1\mathbf{V}_1^\intercal | \cdots | \Sigma_K\mathbf{V}_K^\intercal]$), Procrustes rotation of $\mathbf{X}_k^\intercal\mathbf{U}\Sigma_k$ for $\mathbf{V}_k$, and $\Sigma_k = \mathrm{diag}(\mathbf{U}^\intercal\mathbf{X}_k\mathbf{V}_k)$. OMCOA is therefore the iterated conditional modes (ICM) algorithm [1] associated with this posterior, and the Gibbs sampler is its stochastic counterpart.

Remark 2 (Block weighting)**.** *With block-specific precisions, $\mathbf{F}_\mathbf{U} = \sum_k \tau_k\mathbf{X}_k\mathbf{V}_k\Sigma_k$ weights each block by its estimated noise precision, so that noisier blocks contribute less to the common space. To reproduce exactly the OMCOA criterion* (1)*, a common precision $\tau_1 = \cdots = \tau_K = \tau$ can be used, with $\tau \mid \cdot \sim \mathrm{Gamma}\big(a_0 + \sum_k (N-1)p_k/2,\ b_0 + \sum_k \|\mathbf{X}_k - \mathbf{U}\Sigma_k\mathbf{V}_k^\intercal\|^2/2\big)$.*

Remark 3 (What the sampler adds)**.** *Besides point estimates close to the OMCOA solution, the sampler provides the posterior distribution of the block weights $\sigma_k^{(h)}$ and of the noise levels τ_k^{-1}. Credible intervals on $\sigma_k^{(h)}$ can be used to assess whether a dimension is common to all blocks, shared by some of them, or specific to one block.*

Remark 4 (Practical aspects)**.** *Starting from the OMCOA solution shortens the burn-in. The column-wise sweeps make successive draws of $\mathbf{U}$ and $\mathbf{V}_k$ correlated, so convergence should be checked on trace plots of the $\sigma_k^{(h)}$. Because the posterior is strongly concentrated when the noise is small, the chain rarely jumps between the $H!$ equivalent modes; the alignment step remains necessary nonetheless, in particular when several dimensions have close weights.*

References

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